Ryu Adams

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EDUCATION

University of California, Los Angeles *Los Angeles, CA* Expected: 2027

Computer Science, B.S.

California Academy of Mathematics and Science *Carson, CA* 2023

High School Diploma, Concentration in Computer Science and Engineering

RESEARCH EXPERIENCE

Verifiable & Control-Theoretic Robotics Laboratory (VECTR) Sept. 2024 - Present

UCLA, PI: Brett T. Lopez

- Real-Time Trajectory Planning with Motion Primitive Search Project
  - Developing novel neural controllers to generate real-time optimal trajectories for path planning framework using NeuralODEs
  - Integrated 3D Jump Point Search (JPS) graph search algorithm to be used in real-time motion planning framework, outperforming classical graph search algorithms (A*, Djikstras)
  - Devised monte carlo simulation to compare STITCHER optimal trajectory framework with other state-of-the-art models using ROS1 and RViz

Robotic Decision Making Laboratory (RDML) June - Dec. 2024

Oregon State University, PI: Geoffrey A. Hollinger

- Autonomous Underwater Docking Project
  - Implemented a 3D reconstruction algorithm using opti-acoustic neural rendering (NeRF, AONeuS) to be used in trajectory optimization
  - Designed a ROS package for extracting camera and sonar data from hardware testing
  - Assisted hardware tests at O. H. Hinsdale Wave Research Laboratory
  - Developed an opti-acoustic fiducial marker detection algorithm using Extended Kalman Filter (EKF) for autonomous underwater vehicle (AUV) localization
  - Created pressure sensor data collection pipeline from Raspberry Pi on board BlueROV
  - Conducted literature review on modern 3D reconstruction methods (NeRF, GSplat)
- Tiburon Subsea HAUV Project
  - Designed 3D model of Tiburon Subsea HAUV using Blender
  - Built simulation for Tiburon Subsea HAUV model using Gazebo

LEADERSHIP EXPERIENCE

Bruin Underwater Robotics (BUR) Sept. 2024 - Sept. 2025

Guidance, Navigation, and Control (GNC) Team Member

- Implemented Error State Extended Kalman Filter for robot localization using stereo camera, IMU, and hydrophone sensor fusion

MentorSEAS Mentor *UCLA* Sept. 2025 - Present

- Advised 10 freshman engineering students at UCLA Engineering Welcome Day and held monthly mentoring sessions/socials with students

Section Leader *Online, Stanford Code in Place* Apr. - Jun. 2024, 2025, 2026

- Taught a class of 10 students through weekly guided Python projects, covering material from Stanford's Intro to Python 106A course

AWARDS AND RECOGNITION

- Acknowledgement in "STITCHER: Real-Time Trajectory Planning with Motion Primitive Search," H. J. Levy, B. T. Lopez, *IEEE Transactions on Robotics (T-RO) 2025*
- University of California Leadership Excellence through Advanced Degrees (UC LEADS) Scholar
- 2026 Boston Marathon Qualifier

SKILLS

Technologies: Python, Julia, MATLAB, C++, PyTorch, ROS1/2, Gazebo, Git/GitHub

Languages: English (native), Japanese (native)